Palm Oil Profitability Analysis (Nigeria)

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Introduction

This report analyzes the profitability and investment potential of palm oil production in Nigeria. It focuses on Edo, Ondo, and Cross River between 2020 and 2024.

Objectives

The aim is to understand production costs, yields, revenue, and overall profitability based on available data. The report also highlights key insights on rainfall impact, return on investment (ROI), and payback period to guide farmers, investors, and policymakers in making better decisions.

The primary objective of this analysis is to evaluate the profitability, investment requirements, and financial returns of palm oil production across selected Nigerian states with the goal to:

- Estimate the investment and operating costs for palm oil production.
- Calculate expected revenue, profit, ROI, and payback period for each state.
- Examine how rainfall and climate affect palm oil yields and profitability.
- Compare performance across states to identify the most profitable locations.
- Provide recommendations for improving returns and guiding investment decisions.

Data and Assumptions

The analysis is achieved using proxy data similar to the Data sources below including cost models, plantation yield data, and rainfall records. Key assumptions include average yields per hectare, cost multipliers for labor by state, and palm oil price projections between 2020 and 2024. Inflation is assumed at 5% annually, and discount rate at 12%. Capital expenditure is amortized over the project lifespan of 10 years.

Selected States: Edo, Cross River, and Ondo.

Average palm oil yield: 4 to 5 tons CPO per hectare.

Discount Rate: 0.12

Profit Target: ₦1,000,000,000 annual profit.

Maturity years: 3years.

Price Projection (₦/ton): 2020 - 1.15M, 2021 - 1.30M, 2022 - 1.45M, 2023 - 1.65M, 2024 - 1.85M.

Source	Description	Access Link
FAOSTAT	National palm oil yield, production, area (2010–2024).	https://www.fao.org/faostat/en/#data/QCL
Okumo Oil PLC	Yield and cost Structure benchmark.	https://okomugroup.com
Presco PLC	Industry cost & financial data	https://presco-plc.com
NIFOR	National price and research Data	https://nifor.gov.ng
World Bank Climate Portal	Rainfall and temperature data by state	https://climateknowledgeportal.worldbank.org

Modeling Approach

Data Processing

- Cleaned and standardized state-level yield, cost, and rainfall datasets.
- Extracted and averaged price data from 2020 to 2024.
- Merged datasets into an excel file.

Revenue & Cost Modeling

For each state and year:

$$\text{Revenue per ha} = \text{Yield (tons/ha)} \times \text{Price (₦/ton)}$$
$$\text{Profit per ha} = \text{Revenue/ha} - \text{Operating Capital}/\text{ ha} - \text{Annualised Capital Expenditure/ha}$$

Break-even Analysis

To achieve ₦1,000,000,000 annual profit:

$$\text{Hectares Needed} = \frac{1{,}000{,}000{,}000}{\text{Profit per ha}}$$

ROI & Payback

$$ROI = \frac{\text{Annual Profit}}{\text{Total Investment}} \times 100$$

$$\text{Payback Period} = \frac{\text{Total Investment}}{\text{Annual Profit}}$$

Scenario Modeling

Three profitability scenarios were modeled:

- **Conservative**
- **Base**
- **Optimistic**

Key Insight

Metrics	Edo	Ondo	Cross River
Profit per Hectare #	25.32Million	22.14Million	23.64Million
Average Base Hectare needed	197.56	225.95	211.57
Base Investment Required #	2.37Billion	2.7Billion	2.5Billion
Base Payback Years	0.49	0.55	0.52
Return on Investment %	10.5	9.2	9.8

- Edo and Cross River show the highest profitability per hectare due to superior yield potential and moderate cost structure.
- Ondo has slightly higher Investment amount, which reduce overall margins.
- The Optimistic scenario consistently delivers a 4 to 5% higher ROI compared to the Base case in all state.
- Payback periods range between 2 and 6 years across the three scenarios, depending on yield and price performance of the state.
- The rainfall in Cross River is higher compared to its yield per hectare though has the second-best yield, which indicates taking advantage of high-precipitation regions.
- The project operates at a loss for the year 2020 to 2022 due to high capital and operating costs before 3-year full yield maturity.
- The annual break-even point, where annual profit turns positive happens 2022 and 2023. From 2023, profits grow steadily, reflecting stabilized yields and favorable price assumptions.
- The magnitude of profit increases year-on-year, suggesting strong long-term viability once the plantation matures.

Answering the Requirement needed for #1billion Palm oil Annual Profit

- Scale: Ranges from 137-226 hectares depending on state and scenario. Edo state has the best scale of 137 hectares for the base case and 198 hectares for the conservatives.
- Sustainability: Edo - 9.5/10 suitability, highest profit/ha (₦25M-36M), Cross River - 9.0/10 suitability, Ondo - 8.5/10 suitability
- Investment: ₦7.62 billion total investment required.
- Returns: ROI: 29.62% (base case).
- Payback: 5.96 years.
- Internal Rate Return: 87%.

Recommendations

- Prioritize investments in Edo and Cross River due to their high yield efficiency and strong return metrics.
- Implement phased expansion, starting with 200 to 250 hectares to validate yield and cost assumptions before scaling.
- Monitor rainfall and soil quality as key predictive factors for long-term profitability.
- Reinvest early profits to support mechanization and cost reduction strategies.
- Use conservative price scenarios for financial planning to mitigate commodity volatility risks.